

Collection of Quiz Questions

1 Introduction to Probability

- Let X be a random variable with probability density function (pdf) $p_X(x)$. Which of the following statements are true? (*Select all that apply.*)
 - ☐ The integral of $p_X(x)$ over the whole domain needs to be 1
 - ☐ It holds $p_X(x) \leq 1$ on the whole domain
 - ☐ It holds $p_X(x) = p_X(-x)$ everywhere, so the pdf is symmetric around 0
 - ☐ The cdf $F_X(x)$ can be obtained as the integral $\int_{-\infty}^x p_X(t) dt$
- The posterior can be calculated... (*Select one.*)
 - ☐ By multiplying likelihood and prior
 - ☐ By dividing the likelihood by the evidence
 - ☐ By multiplying likelihood and prior, and dividing by the evidence
 - ☐ By multiplying evidence and prior
- The likelihood (*Select one.*)
 - ☐ is used to specify how probable we think the different hypotheses are
 - ☐ is used to compute how probable the data is under the different hypotheses
- A classical application of Bayes' theorem is medical diagnosis. Suppose if someone is sick, than the test correctly identifies it with a probability of 99%, i.e. $\Pr(\text{positive} \mid \text{sick}) = 0.99$. If someone tests positive, what is the probability they actually have the disease? (*Select one.*)
 - ☐ 0.99
 - ☐ If the test is positive, the patient is certain to have the disease, so 1
 - ☐ We don't have enough knowledge to calculate it
- In a medical setting, assume the following: Patients can be sick or healthy; there is a test for the disease that can be positive or negative. Assuming that we have knowledge about the true probabilities in Table 1, *select all that are true.*
 - ☐ $\Pr(\text{positive}) = 0.11$
 - ☐ $\Pr(\text{positive}) = 0.01$
 - ☐ $\Pr(\text{sick} \mid \text{positive}) = \frac{0.01}{0.11} = \frac{1}{11}$

	Test positive	Test negative	Marginal
Sick	0.01	0.00	0.01
Healthy	0.10	0.89	0.99
Marginal	0.11	0.89	

Table 1: Caption

$$\square \Pr(\text{sick} \mid \text{positive}) = \frac{0.01}{0.01} = 1$$

6. Central limit theorem: Let $X_1, \dots, X_n$ be i.i.d. random variables from the same distribution with mean zero and finite variance σ^2 . Let $Z = \frac{1}{\sqrt{n}} \sum_{i=1}^n X_i$. (Select all that apply.)

- ☐ The distribution of Z tends to the Gaussian distribution as n increases if and only if the probability distribution of $X_1, \dots, X_n$ is Gaussian, uniform or exponential
- ☐ The distribution of Z tends to the Gaussian distribution as n increases for any arbitrary probability distribution
- ☐ For the theorem to hold, n must be at least 1000, since small values of n are never sufficient for the approximation to be valid
- ☐ The mean of Z is zero

2 Generative Models for Discrete Data

1. Imagine you are playing the number game, drawing independently and with replacement numbers from 1 to 100. In what follows, $\mathcal{D} = \{x_1, \dots, x_n\}$ is the observed data, $\mathcal{H}$ the space of all hypotheses (in this case, arithmetic concepts) and $h \in \mathcal{H}$ an individual hypothesis.

- (a) You observe one number, $\mathcal{D} = \{9\}$. Which of the following hypotheses has the highest likelihood? Select one.

- ☐ $h_1 = \{\text{Odd numbers}\}$
- ☐ $h_2 = \{\text{Multiples of 3}\}$
- ☐ $h_3 = \{\text{only 9}\}$
- ☐ $h_4 = \{\text{Even numbers}\}$

- (b) Your new observations are $\mathcal{D} = \{9, 36, 6, 81\}$. Which hypotheses from the previous question remain valid? Select all that apply.

- ☐ $h_1 = \{\text{Odd numbers}\}$
- ☐ $h_2 = \{\text{Multiples of 3}\}$
- ☐ $h_3 = \{\text{only 9}\}$
- ☐ $h_4 = \{\text{Even numbers}\}$

- (c) Artificial hypotheses such as “only 9” or “powers of two without 32” tend to have a high likelihood. How can you still favor simpler, more natural hypotheses? Select one.

- ☐ picking an appropriate prior
- ☐ calculating the posterior predictive

- (d) The likelihood for drawing the first sample x_1 given a hypothesis h is $P(x_1|h) = \frac{1}{|h|}$, where $|h|$ is the count of elements in our hypothesis. Which assumption is important to arrive at this model? Select one.

- ☐ that the prior is uniform over all hypotheses
- ☐ that we sample with replacement
- ☐ that we sample numbers uniformly from within the hypothesis

- (e) When drawing n data points, the joint likelihood is $P(\mathcal{D} | h) = \left(\frac{1}{|h|}\right)^n$. Which assumptions are important to arrive at this likelihood? Select all that apply.

- ☐ each sample depends on the previous one
- ☐ each sample is drawn independently
- ☐ samples are drawn with replacement

- ☐ samples are drawn without replacement
2. Consider a random variable distributed according to a Bernoulli distribution, $X \sim \text{Ber}(\theta)$, with $\theta \in [0, 1]$. You want to estimate the true value of θ with a Bayesian approach: which of the following distributions are valid priors for θ ? Select all that apply.
- ☐ The uniform distribution, $\theta \sim \text{Unif}(0, 1)$
 - ☐ The standard normal distribution, $\theta \sim \mathcal{N}(0, 1)$
 - ☐ A generic normal distribution, $\theta \sim \mathcal{N}(\mu, \sigma^2)$
 - ☐ The Beta distribution, $\theta \sim \text{Beta}(a, b) \propto \theta^{a-1}(1 - \theta)^{b-1}$

3 Regression

In regression, you are trying to predict a target noisy function of some input, $y = f(\mathbf{x}) + \eta$.

1. Suppose that the function is linear, $f(\mathbf{x}) = \mathbf{w}^\top \mathbf{x}$, and the noise is Gaussian, $\eta \sim \mathcal{N}(0, \sigma^2)$.
- (a) How is y distributed?
- ☐ We don't have enough information
 - ☐ Like the original, possibly unknown data distribution $p(\mathbf{x})$
 - ☐ Like a Gaussian distribution with mean $\mathbf{w}^\top \mathbf{x}$ and variance σ^2 , $\mathcal{N}(\mathbf{w}^\top \mathbf{x}, \sigma^2)$
 - ☐ Like a standard normal distribution, $\mathcal{N}(0, 1)$
- (b) Consider a dataset of input-observation pairs, $\mathcal{D} = \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n)\}$, and say you can write the likelihood for $\mathbf{w}$ as

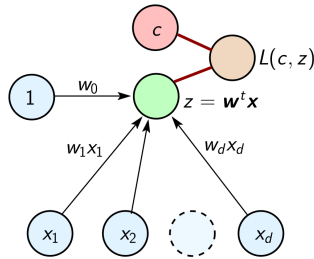
$$L(\mathbf{w}) \propto \prod_i \exp\left[-\frac{1}{2\sigma^2}(y_i - \mathbf{w}^\top \mathbf{x}_i)^2\right].$$

Which additional assumptions have been made? (*Select all that apply*)

- ☐ The observations are independent
 - ☐ The inputs are Gaussian distributed
 - ☐ The noise distribution is a standard normal
 - ☐ The observations y_i are identically distributed
2. In our Bayesian linear regression model, we use the same likelihood as above and add a prior on the model weights $p(\mathbf{w}) = \mathcal{N}(\mathbf{w}|0, \tau^2 \mathbf{I})$.
- (a) This means that the model parameter $\mathbf{w}$ is treated as a random variable.
- ☐ True
 - ☐ False
- (b) This prior encourages small values of $\mathbf{w}$: how can we make this preferences stronger?
- ☐ It is not possible under this model
 - ☐ Making τ^2 larger
 - ☐ Making τ^2 smaller
- (c) The resulting posterior is $p(\mathbf{w}|\mathcal{D})$. (*select all that apply*)
- ☐ The posterior follows a uniform distribution
 - ☐ The posterior follows a Gaussian distribution
 - ☐ Sampling from the posterior means sampling a model
 - ☐ Sampling from the posterior generates new plausible observations

4 Classification

- The single-sample perceptron algorithm has the following properties (*select all that apply*)
 - ☐ it converges for linearly separable data
 - ☐ it does not necessarily converge for data that is not linearly separable
 - ☐ the order of data points decides if it converges or not
 - ☐ it does not guarantee a high separation margin
- Discriminative classifiers first define the generative process with class densities before deriving the discriminative function from the class densities.
 - ☐ True
 - ☐ False



- If the loss $L(c, z)$ takes the form

$$L_i = \sum_{i=1}^n c_i \log \sigma(\mathbf{w}^t \mathbf{x}_i) + (1 - c_i) \log(1 - \sigma(\mathbf{w}^t \mathbf{x}_i)),$$

which model is represented in this computational graph?

- ☐ the perceptron
 - ☐ logistic regression
- Consider the sigmoid function $\sigma(x)$. Which of the following properties are true? (*Select all that apply*)
 - ☐ It models the class probability of Bernoulli random variables
 - ☐ It holds $\sigma(-x) = 1 - \sigma(x)$
 - ☐ It maps the real numbers to the unit interval, i.e. $\sigma: \mathbb{R} \mapsto [0, 1]$
 - ☐ It has a discontinuity in $x = 0$
 - In a logistic regression model for a binary classification problem, we can train the parameters $\mathbf{w}$ by applying the Gradient Descent (GD) algorithm to the negative log-likelihood, leading to the (online) update rule

$$\mathbf{w} \leftarrow \mathbf{w} + \eta(c_i - \pi_i)\mathbf{x}_i, \quad \pi_i = P(c_i = 1 \mid \mathbf{x}_i, \mathbf{w}). \quad (1)$$

How can we recover the perceptron algorithm from it? (*Single choice*)

- ☐ By setting the learning rate η to 1
- ☐ By approximating the probability π_i with the most probable label
- ☐ By imposing a Gaussian prior on the weights, $p(\mathbf{w}) = \mathcal{N}(\mathbf{w} \mid \mu, \sigma^2)$
- ☐ By using the analytical maximum of the likelihood instead of approximating it with iterative techniques

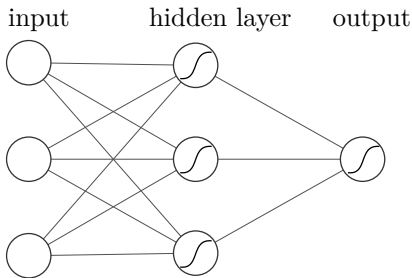
5 Neural Networks

- Linear models are based on linear combinations of the features, potentially transformed by basis functions $\phi_j(\mathbf{x})$, e.g. the polynomial basis $\phi_j = x^j$:

$$y(\mathbf{x}, \mathbf{w}) = f\left(\sum_{j=1}^d w_j \phi_j(\mathbf{x})\right), \quad (2)$$

where $f(\cdot)$ is the identity for regression or a suitable activation function for classification. Interestingly, neural networks can be written in the same way. However, out of the following statements, which one is only true for neural networks and not for linear models? (*Single choice*)

- ☐ The basis functions ϕ_j can be nonlinear
- ☐ There can be a bias term, e.g. $\phi_0(\mathbf{x}) = \text{const}$
- ☐ The model can be trained via gradient descent
- ☐ The basis functions ϕ_j are not fixed but learnable



- Consider a neural network with one hidden layer and a scalar output. The sensitivity of the output unit is δ_k . On which of the following quantities does δ_k depend?
 - ☐ the weights of the output unit
 - ☐ the activation function of the output unit
 - ☐ the sensitivities of units in the hidden layer
- Consider a neural network with one output neuron, that has the activation function $z = \sin(a)$, where a is the neuron's net activation. With MSE as the loss function,

$$J = \frac{1}{2}(t - z)^2, \quad (3)$$

what is the sensitivity of the output neuron? (*Single choice*)

- ☐ $-(t - z) \cos(a)$
- ☐ $-(t - z) \sin(a)$
- ☐ J

- Let $f: [0, 1]^d \rightarrow \mathbb{R}$ be a continuous function on the d -dimensional hypercube. Which of the following statements are true? (*Multiple choice*)
 - ☐ Kolmogorov-Arnold-Lorentz theorem provides a way of finding simple (i.e. single-variable) functions whose superposition can represent f
 - ☐ f can be approximated arbitrarily well by polynomials

- ☐ f can be approximated arbitrarily well by a neural network with only linear activations, provided the hidden layer is wide enough
 - ☐ f can be approximated arbitrarily well by a neural network with only ReLU activations, provided the hidden layer is wide enough
5. For a deep model with the number of hidden layers L and the number of neurons per hidden layer m , the number of linear regions grows (*Multiple choice*)
- ☐ polynomially in m
 - ☐ exponentially in m
 - ☐ polynomially in L
 - ☐ exponentially in L

6 Statistical Learning Theory

1. In Statistical Learning Theory (SLT), given some data $\mathbf{x} \sim p^*(\mathbf{x})$ and the corresponding labels $y \sim p^*(y | \mathbf{x})$, the goal is to find a function $f(\mathbf{x})$ which predicts labels on unseen data in the best possible way. The *expected risk* is defined as

$$R[f, p^*] = \mathbb{E}_{(\mathbf{x}, y) \sim p^*(\mathbf{x}, y)} [\mathcal{L}(y, f(\mathbf{x}))], \quad (4)$$

where $\mathcal{L}(\cdot)$ is an appropriate loss function. Which of the following statements are true in general? (*Select all that apply*)

- ☐ Minimizing the expected risk does not guarantee to find a good function, so we introduce the *empirical risk* $R_{\text{emp}}[f]$
 - ☐ We cannot minimize $R[f, p^*]$ since we don't have access to the joint distribution $p^*(\mathbf{x}, y)$
 - ☐ If we restrict ourselves to an appropriate hypothesis space $\mathcal{H}$, then we can minimize $R[f, p^*]$ directly
 - ☐ The empirical risk $R_{\text{emp}}[f]$ is defined as the sample average of the loss over a dataset $\mathcal{D}$
2. Consider a binary classification problem, i.e. $y \in \{-1, +1\}$. Let f^* be the best possible learnable function,

$$f^* = \arg \min_{f \in \mathcal{H}} R[f]. \quad (5)$$

Given a dataset $\mathcal{D} = \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n)\}$, consider the function

$$f_n(\mathbf{x}) = \begin{cases} f^*(\mathbf{x}) & \mathbf{x} \in \mathcal{D}, \\ -f^*(\mathbf{x}) & \text{otherwise.} \end{cases} \quad (6)$$

What can you say about f_n ? (*Single choice*)

- ☐ f_n minimizes both the empirical and the expected risk
 - ☐ f_n minimizes only the empirical risk
 - ☐ f_n minimizes only the expected risk
 - ☐ f_n does not minimize neither the expected nor the empirical risk
3. Let f^* be the best possible learnable function as before, and f_n an empirical risk minimizer for a dataset $\mathcal{D}$. Convergence in the probability of which quantities is required for the principle of Empirical Risk Minimization (ERM) to be consistent? (*Multiple choice*)
- ☐ $R_{\text{emp}}[f_n]$ and $R[f^*]$
 - ☐ $R[f_n]$ and $R[f^*]$
 - ☐ The ERM principle is always consistent

☐ $R_{\text{emp}}[f_n]$ and $R[f_n]$

4. Which of the following statements are true? (*Multiple choice*)

☐ Linear classifiers in $\mathbb{R}^d$ have VC dimension d

☐ The VC dimension of a class of functions is equal to the number of free parameters

☐ A function class parametrized by a single real parameter $\theta \in \mathbb{R}$ cannot have infinite VC dimension

☐ The function $\text{sgn}(\cos(\alpha x))$, $\alpha \in \mathbb{R}$ can perfectly classify any number of points

7 Support Vector Machines

1. In the statistical learning theory chapter we learned that having larger margin means (multiple choice)

☐ shrinking the upper bound of the VC dimension of decision functions

☐ tightening the upper bound of the true risk

☐ reducing the norm of the decision plane weight vector $\|\mathbf{w}\|$

☐ a larger training set is no longer needed to reduce the true risk

2. In order find the solution for SVMs, we have seen the optimization problem

$$\min_{\mathbf{w}} \left[\sup_{\alpha_i \geq 0} \left(\frac{\|\mathbf{w}\|^2}{2} + \sum_{i=1}^n \alpha_i (1 - y_i [\mathbf{x}_i^t \mathbf{w}]) \right) \right] \quad (7)$$

(a) Which of the following statements about it are true? (multiple choice)

☐ Strong duality holds

☐ Slater's condition holds

☐ It is convex in $\mathbf{w}$

☐ It is non-convex in $\mathbf{w}$

(b) How did we solve the aforementioned optimization problem? (single choice)

☐ Swapping the min and the sup operation and analytically minimizing over $\mathbf{w}$ to obtain the dual problem over α_i

☐ Setting all α_i to zero such that they fulfill the constraint and numerically minimizing $\mathbf{w}$

☐ Swapping the min and the sup operation and using gradient descent to optimize $\mathbf{w}$ and α_i simultaneously to solve the dual problem

☐ Setting all α_i to zero such that they fulfill the constraint and using gradient descent to optimize over $\mathbf{w}$

Which interesting properties did the solution of the dual problem exhibit? (multiple choice)

☐ The solution depends only on the support vectors

☐ The solution depends on all training points equally

☐ The solution depends only on inner products of training points

8 Gaussian Processes

1. A Gaussian process $\mathcal{GP}$ is often described as a distribution over functions.

(a) Which of the following statements are true? (*Select all that apply*)

☐ A Gaussian process is fully specified by a mean function and a covariance function

☐ A Gaussian process samples functions from a fixed finite-dimensional parametric family

- ☐ For any finite set of input points, the corresponding function values are jointly Gaussian distributed
 - ☐ A Gaussian process assigns independent Gaussian distributions to $f(x)$ at every input x
- (b) Let $f(x) \sim \mathcal{GP}(m(x), k(x, x'))$. What does the covariance function $k(x, x')$ describe? (*Single choice*)
- ☐ The probability that two inputs x and x' are equal
 - ☐ The expected value of the function at input x
 - ☐ The amount of independent observation noise at input x
 - ☐ How strongly the function values $f(x)$ and $f(x')$ vary together
2. Gaussian processes rely on important properties of the Gaussian distribution. Which property is used when deriving the posterior distribution over test function values? (*Single choice*)
- ☐ A covariance matrix is diagonal
 - ☐ Gaussian distributions describe independent random variables
 - ☐ The conditional distribution of a Gaussian is always a Gaussian
 - ☐ The product of two Gaussians is always a Gaussian
3. A covariance function $k(x, x')$ defines the entries of the covariance matrix used by the Gaussian process. Which condition must a valid covariance matrix satisfy? (Select all that apply.)
- ☐ It must be symmetric
 - ☐ It must be positive semi-definite
 - ☐ It must be diagonal
 - ☐ Its entries must be between 0 and 1
4. Consider the Gaussian RBF covariance function

$$k(x, x') = \exp\left(-\frac{1}{2l^2} \|x - x'\|^2\right). \quad (8)$$

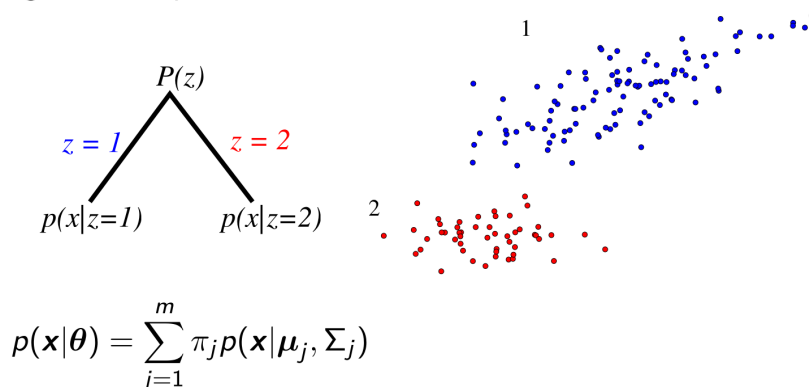
What is the qualitative effect of increasing the length scale l ? (Select one.)

- ☐ Prior functions become less smooth and vary more rapidly
- ☐ Prior functions become smoother and vary more slowly
- ☐ The covariance matrix becomes independent of the inputs

9 Gaussian Mixture Models

1. Assume our model is a mixture of Gaussians as seen in the lecture (see Figure). (Select all that apply.)
- ☐ Every data point x could have been generated in two ways
 - ☐ Every data point x could have been generated in only one way
 - ☐ To generate a data point, we sample from two Gaussians and make a weighted sum with weight π_j
 - ☐ To generate a data point, we sample from only one Gaussian with probability π_j
2. What does the E-step of the EM algorithm compute?
- ☐ New means and covariances for each Gaussian
 - ☐ The maximum likelihood parameters given hard cluster assignments

Data generation process:



- ☐ Posterior probabilities $P(z = j \mid x_i, \theta^t)$ for each data point and component
 - ☐ The BIC score for model selection
3. In the M-step for a Gaussian Mixture Model, the updated mean $\hat{\mu}_j$ is computed as:
- ☐ The sample mean of all data points
 - ☐ The sample mean of data points hard-assigned to component j
 - ☐ A weighted mean of all data points, weighted by their soft assignment to component j
 - ☐ The median of data points assigned to component j
4. In deriving the EM lower bound, Jensen's inequality is applied. Which form is the correct one?
- ☐ $\log(\mathbb{E}[X]) \leq \mathbb{E}[\log(X)]$
 - ☐ $\log(\mathbb{E}[X]) \geq \mathbb{E}[\log(X)]$
5. What are differences between a standard Gaussian Mixture Model and a Mixture of Experts model?
- ☐ Only Mixture of Experts use the EM algorithm
 - ☐ In Mixture of Experts, the mixing weights depend on the input x , not just a global prior
 - ☐ Standard GMMs can have more than two components; Mixture of Experts cannot
 - ☐ Mixture of Experts models do not have latent variables